Practical impacts: bounded mechanism, measurable indicators, and safeguards

Potential SDG connections. Current evidence remains a stylized computational benchmark.

RESEARCH MECHANISM

CCAR conditions access to enhanced non-safety coordination information on voluntary M/P/F

Mandatory safety information remains unconditional in every modeled state.

AFFECTED STAKEHOLDERS

Commercial operators
Traffic-management authority
Customers and nearby communities
Emergency and public-safety users

Payment-based commercial priority does not govern emergency/public-safety missions.

INDICATOR AND BASELINE

Proposed pilot indicator: mean coordination delay and type-specific enhanced-access rate under CCAR versus baseline.

Current modeled comparator: underlying welfare gap at Medium congestion, 0.946 baseline versus 0.050 CCAR.

No field delay, safety, throughput, emissions, or equity outcome has been measured.



SDG TARGET 9.1 POTENTIAL

Quality, reliable, sustainable and resilient infrastructure, with affordable and equitable access.

Connection: auditable coordination rules for shared infrastructure.
Missing evidence: reliability, affordability, and access outcomes.
Risk: high-cost operators receive systematically less enhanced access.
Safeguard: type-blind safety floor, access audit, appeal, and suspension rule.



SDG TARGET 11.2 POTENTIAL

Safe, affordable, accessible and sustainable transport systems for all.

Connection: possible future support for urban logistics coordination.
Missing evidence: public access, safety, and sustainability effects.
Risk: commercial efficiency crowds out vulnerable or public missions.
Safeguard: protected public-safety priority and public accountability.

ACADEMIC STANDARD

Reproducible equilibrium and robustness
Claims bounded to modeled outcomes

INDUSTRY STANDARD UNVERIFIED

Latency, reliability, security, accuracy
Pilot calibration and implementation cost

PUBLIC STANDARD UNVERIFIED

Safety, equitable access, transparency
Independent oversight and correction

Official targets: https://sdgs.un.org/goals/goal9 and /goal11. Official icons: UN communications materials, English web set.

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